

Project Report: AI-Powered Credit Card Fraud Detection Pipeline

A Comprehensive Machine Learning Approach using Ensemble Methods
and SHAP Interpretability

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Abstract

Financial fraud is a rapidly growing menace in the digital economy, with credit card fraud accounting for billions of dollars in losses annually. Traditional, rule-based detection systems struggle to adapt to the highly dynamic and complex patterns of modern fraudulent behavior. In this project, we propose and develop an end-to-end Machine Learning pipeline designed to detect fraudulent credit card transactions in real-time. Utilizing a highly imbalanced European credit card dataset, this research mitigates extreme class skewness using the Synthetic Minority Over-sampling Technique (SMOTE). Four classifiers were evaluated: Logistic Regression, Linear SVC, Random Forest, and Extreme Gradient Boosting (XGBoost). The XGBoost classifier demonstrated superior performance in capturing anomalous transactional patterns. To resolve the "black-box" dilemma of complex ensemble models, SHapley Additive exPlanations (SHAP) is integrated to provide mathematical interpretability for every decision. Finally, the system is deployed via an interactive Streamlit web dashboard for real-time risk assessment and visualization.

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1 Introduction

With the exponential rise of e-commerce and digital banking, credit cards have become the primary mechanism for international and domestic transactions. Consequently, they have also become the primary target for malicious financial exploitation, with cybercrime costing the global economy trillions of dollars [2]. The central objective of this project is to construct a scalable, highly accurate Machine Learning pipeline capable of distinguishing between legitimate and fraudulent transactions.

Rather than stopping at localized model training, this project bridges the gap between data science and software engineering by creating a full-stack application. It implements robust Exploratory Data Analysis (EDA) using standard Python data structures [14], handles extreme statistical distribution flaws, optimizes high-performance predictive models using the Scikit-Learn framework [15], guarantees algorithmic transparency, and delivers the final engine through a modern User Interface (UI).

2 Motivation

Financial institutions are under immense pressure to detect fraud instantly to prevent devastating monetary loss and protect customer trust. However, extensive surveys of fraud detection systems [1] indicate that traditional systems rely excessively on static, manually created rules that fail to capture the evolving, sophisticated tactics of modern cybercriminals. This project is motivated by the urgent need for a dynamic, self-learning Artificial Intelligence system that can adapt to new fraud patterns autonomously. Furthermore, regulatory frameworks in the financial sector demand strict transparency; therefore, integrating Explainable AI (XAI) techniques to understand exactly *why* an algorithm blocked a transaction is critical.

3 Objectives

The primary objectives of this project are formally defined as follows:

- **Data Profiling:** To perform comprehensive Exploratory Data Analysis (EDA) on a high-dimensional financial dataset to identify critical transactional patterns.
- **Imbalance Mitigation:** To mathematically resolve extreme class imbalance utilizing the Synthetic Minority Over-sampling Technique (SMOTE).
- **Model Optimization:** To design, train, and comparatively evaluate multiple machine learning classifiers, identifying the optimal model for tabular anomaly detection.
- **Algorithmic Transparency:** To integrate SHapley Additive exPlanations (SHAP) to eliminate the "black-box" nature of ensemble models and provide actionable interpretability.

- **System Deployment:** To deploy the final predictive engine into an interactive frontend dashboard for real-time transaction screening.

4 Literature Review

Detecting credit card fraud is fundamentally an anomaly detection problem. Historically, banks relied on hard-coded rulesets, which produced high false-positive rates. Bhattacharyya et al. [3] conducted foundational comparative studies demonstrating the clear superiority of advanced data mining and machine learning techniques over these traditional methods for predicting anomalous purchasing behavior.

However, modern fraud datasets severely suffer from extreme class imbalance, a well-documented roadblock in machine learning [10], often leading algorithms to incorrectly predict the majority class. Chawla et al. [?] addressed this discrepancy by introducing SMOTE, a technique that synthesizes minority class examples rather than simply duplicating them, vastly improving decision boundary mapping.

For classification tasks, foundational models such as Logistic Regression [11] and Support-Vector Networks [6] have been historically prevalent. However, ensemble methods like Random Forests [4] and Gradient Boosting [8] have proven highly effective for complex tabular data. Xuan et al. [18] explicitly demonstrated that tree-based architectures excel at capturing the non-linear intricacies of credit card transaction data. Building on this, Chen and Guestrin [17] formalized XGBoost, a scalable tree boosting system that minimizes over-fitting while maximizing computational execution speed—making it an ideal engine for millisecond-latency financial systems. Finally, to ensure algorithmic trust, Lundberg and Lee [?] introduced SHAP, a unified framework derived from cooperative game theory to accurately interpret and explain machine learning predictions.

5 Problem Definition

The problem is formalized as a binary classification task. Given a transaction X_i containing n features (amount, time elapsed, and 28 anonymized principal components), the objective is to learn a mapping function $f(X_i) \rightarrow y_i$, where $y_i \in \{0, 1\}$.

- $y_i = 0$ denotes a *Legitimate* transaction.
- $y_i = 1$ denotes a *Fraudulent* transaction.

The primary challenge is maximizing the Detection Rate (Recall) of $y_i = 1$ without unacceptably increasing the False Positive Rate.

6 Proposed System Architecture

The system architecture follows a linear, highly decoupled pipeline:

1. **Data Ingestion:** Loading the raw CSV dataset securely.
2. **Preprocessing Strategy:** Standardizing non-PCA features (Time, Amount) using Z -score normalization.
3. **Resampling Engine:** Applying SMOTE to the training subset to achieve a 1 : 1 class distribution.
4. **Training Engine:** Iterating through Logistic Regression, Linear SVC, Random Forest, and XGBoost.
5. **Explainability Layer:** Interfacing the serialized best model (XGBoost) with SHAP TreeExplainers.
6. **Deployment Dashboard:** Exposing the pipeline via a Streamlit front-end.

7 Dataset & Exploratory Data Analysis (EDA)

The dataset utilized is the European Credit Card Fraud Dataset [19]. It contains 284,807 transactions executed over 48 hours in September 2013.

7.1 Feature Space

To protect consumer privacy, the original features were transformed using Principal Component Analysis (PCA)—a well-established dimensional reduction technique [12]—resulting in 28 continuous numerical features ($V1$ through $V28$). The only untouched features are *Time* and *Amount*.

7.2 Class Imbalance Analysis

Exploratory Data Analysis revealed a severe skewness. Out of 284,807 total transactions, only 492 were fraudulent. This constitutes a mere **0.172%** anomaly rate. Without mathematical intervention, a naïve classifier would achieve 99.8% accuracy by blindly predicting 0 for every input, completely failing its primary objective.

8 Methodology

8.1 Data Preprocessing

Because algorithms and gradient-descent optimizers are highly sensitive to feature magnitudes, we applied a *StandardScaler* to the *Amount* and *Time* features:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

A separate scaler object was saved locally (`scaler.pkl`) to apply identical transformations to live inputs during deployment.

8.2 Algorithmic Balancing via SMOTE

To resolve the 0.172% imbalance, the dataset was split into training (80%) and testing (20%) sets. **SMOTE** was applied exclusively to the training set to prevent data leakage. SMOTE operates by finding the k -nearest neighbors of a minority class sample and mathematically generating synthetic data points along those vectors, achieving a perfect 50/50 distribution for the algorithms.

8.3 Model Selection

Four models were instantiated utilizing Scikit-Learn [15] and XGBoost libraries:

- **Logistic Regression:** A probabilistic discriminative model [11].
- **Linear SVC:** Support Vector Classifier optimized for high-dimensional margins [6].
- **Random Forest:** An ensemble of decision trees utilizing bagging [4].
- **XGBoost:** Extreme Gradient Boosting [17], configured to utilize GPU acceleration.

9 Results and Discussion

Because overall Accuracy is deceptive in anomalous datasets, the models were evaluated strictly using confusion-matrix derived metrics optimized for imbalance, particularly focusing on the Precision-Recall relationship [7]:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

9.1 Model Performance Analysis

While Logistic Regression and Linear SVC struggled to define complex non-linear boundaries, the tree-based ensemble methods performed exceptionally well. **XGBoost** outperformed all competitors, capturing the highest metric of legitimate fraud attempts (Recall) while heavily penalizing false positives to maintain a high F1-Score. The trained XGBoost model was ultimately selected and serialized as `best_model.pkl` for the production environment.

10 Model Interpretability (SHAP)

A major operational requirement in Fintech is explainability. If a customer's transaction is blocked by an AI, the institution must be able to mathematically justify the decision. The XGBoost framework was augmented with **SHAP** [?]. Using game-theoretic Shapley values, the pipeline calculates the exact proportional contribution of every feature ($V1...V28$, Amount, Time) to the final algorithmic probability score. This allows the system to generate visual Waterfall Plots displaying exactly which variables triggered a specific Fraud Alert.

11 Deployment & User Interface

The entire machine learning pipeline was abstracted behind a fully interactive web application using the **Streamlit** library [20]. Applying modern MLOps principles [16], the dashboard allows end-users to input simulated transactions by extracting random normal or fraudulent rows directly from the raw database, or manually dial in custom variables for sandbox testing.

12 Advantages and Limitations

12.1 Advantages

- **High Detection Rate:** The inclusion of SMOTE ensures the model is hyper-aware of fraudulent patterns that standard baseline models miss.
- **Strict Transparency:** SHAP integration provides a rigorous, mathematical explanation for every single transaction denied by the system, ensuring trust and regulatory compliance.

12.2 Limitations

- **Concept Drift:** Fraud patterns evolve over time as criminals adapt. The model will require continuous retraining on new data streams to maintain its defensive edge.
- **PCA Obfuscation:** Because the dataset features ($V1 - V28$) were transformed via PCA [12] by the original authors to protect user privacy, it is difficult to map the SHAP explanations back to tangible, real-world variables.

13 Conclusion & Future Work

This project successfully conceptualized, trained, and deployed an end-to-end Machine Learning ecosystem for Credit Card Fraud Detection. By addressing the severe limitations

of imbalanced data through SMOTE, optimizing an XGBoost classifier for peak detection metrics, and removing the "black-box" stigma via SHAP, the system presents a robust, transparent, and highly accurate financial security solution.

Future Work includes deploying the Streamlit application to a cloud infrastructure to test latency bounds under high concurrency. Furthermore, we intend to explore Deep Learning architectures [9], specifically Autoencoders, for unsupervised anomaly detection on unclassified data streams.

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